

Priyanshu Kumar Sharma

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OVERVIEW

Cloud and DevOps-focused software engineer intern with a strong research-driven mindset, experienced in building and evaluating **serverless and multi-cloud systems**. Worked on **cloud performance benchmarking, FaaS platforms, and distributed architectures** across AWS and Azure. Combines backend engineering skills in **Python and Node.js** with DevOps practices to design scalable, automated, and production-ready cloud solutions. Curious problem-solver with a passion for modern cloud platforms and emerging infrastructure technologies. Motivated to build resilient, observable, and scalable cloud platforms.

EDUCATION

Ajeenkya D Y Patil University, Pune

Aug 2022 - Present

Bachelor of Technology, Information Technology

- **GPA:** SCGPA: 9.9 (SEM-7) | CGPA: 9.9
- **Coursework:** Cloud Architecture Design, Network Security Labs, DevOps Pipelines

Sri Chaitanya Jr. Kalasala, Hyderabad

Apr 2020 - May 2022

Intermediate Education

- **Percentage:** 93.9

Sri Chaitanya High School, Hyderabad

Apr 2019 - Mar 2020

Higher Secondary Education

- **Percentage:** 87.6

EXPERIENCE

Marquardt India Pvt. Ltd.

Jun 2025 - Sep 2025

Web Dev Intern

Pune

- Contributed to the development and operational support of internal web applications, gaining exposure to application deployment pipelines, environment setup, and runtime monitoring.
- Supported backend services built with Express.js and MySQL, focusing on performance tuning, error handling, and production readiness.
- Collaborating with senior developers using Git/GitHub to perform unit testing, debugging, and performance tuning of web applications, with a focus on cross-browser compatibility and optimized load times.
- Implemented Jenkins-based CI/CD pipelines to automate Docker image builds and deployments, improving release speed and deployment reliability for internal web services.

Seamless Education and Services (Seamedu) Pvt. Ltd.

Mar 2025 - Jun 2025

IT Intern– Placement Coordinator, IT Department

Pune

- Selected as a Trainee Intern in the Information Technology Department at Seamedu School of Pro-Expressionism, with responsibilities encompassing IT infrastructure exposure, software tools utilization, and process optimization to support departmental objectives.
- Coordinating end-to-end placement activities for the IT Department, including scheduling recruitment drives, managing candidate databases, preparing technical reports, and tracking hiring metrics.
- Facilitating structured communication between students, faculty, and industry recruiters by leveraging digital collaboration platforms, cloud-based data sharing tools, and automated notification systems, demonstrating excellent communication skills to work cross-functionally and across departments.
- Supporting the integration of IT solutions into placement processes to improve operational efficiency, data accuracy, and stakeholder engagement.

Indian Institute of Technology Ropar, Punjab

May 2024 - Jul 2024

Cloud Research Intern

Rupnagar, Punjab

- Engaged in advanced research initiatives under the mentorship of an IIT professor, focusing on distributed computing and serverless architectures.
- Implemented and validated **XFBench and XFaaS frameworks** for benchmarking **Function-as-a-Service (FaaS)** platforms, including end-to-end setup, configuration, and experimental evaluation.
- Designed and deployed **serverless functions on cloud-native platforms**, emphasizing **scalability, modular design, and low-latency execution**.
- Conducting in-depth performance profiling, bottleneck analysis, and optimization of serverless workloads, leveraging benchmarking metrics, instrumentation tools, and workload simulation techniques.
- Selected for the prestigious **Summer Research Fellowship Program (IAS, 2024)**, gaining hands-on exposure to cutting-edge cloud computing research and performance evaluation methodologies.

PROJECTS

Multi-Cloud SaaS Orchestration Platform

Built and deployed a cloud-native multi-cloud orchestration platform with containerized microservices using Docker and infrastructure provisioning via Terraform. Implemented automated background jobs with Celery and Redis for periodic synchronization of cloud resource data from AWS, Azure, and GCP APIs. Designed CI/CD-friendly architecture and production-ready deployments with secure JWT authentication, PostgreSQL-backed services, and scalable FastAPI backend. This project highlights practical experience in infrastructure automation, containerization, service orchestration, and operational reliability. [GitHub](#)

Storage SaaS Platform

I built a **Software-as-a-Service (SaaS) platform** leveraging AI to provide automated business solutions for modern video management. The platform utilizes a robust tech stack, including **Next.js 14** for the frontend framework, **TypeScript** and **JavaScript** for application logic, and **Tailwind CSS** for responsive styling. For secure user authentication, it integrates **Clerk**, while **Prisma ORM** with **Neon DB (PostgreSQL)** manages the database. **Cloudinary** is used for scalable video storage and optimized delivery, and the platform is deployed seamlessly on **Vercel**. AI is harnessed to automate business workflows and enhance video-related operations, creating an efficient, scalable, and user-friendly solution. [GitHub](#)

Quantum Cloud Integration

I developed a **hybrid cloud-quantum system** that seamlessly integrates classical cloud computing with quantum processing to achieve enhanced computational efficiency. The workflow enables interoperability between traditional cloud services and quantum systems, providing secure and scalable solutions. This hybrid architecture is implemented using **AWS** for cloud infrastructure, **IBM Quantum** for quantum resources, and **Docker** for containerization, ensuring flexible deployment and robust integration across platforms. [GitHub](#)

Rural Gyan Platform

I developed a comprehensive **full-stack educational management system** featuring role-based dashboards, integrated AI-powered tools, and bilingual (English/Hindi) support. The platform offers streamlined admin, teacher, and student workflows, including live virtual classes, automated quiz grading, real-time analytics, AI-assisted tutoring, and activity monitoring. The tech stack includes **React.js** and **Tailwind CSS** for the frontend, **Node.js**, **Express.js**, and **Socket.io** for the backend, and **MongoDB** for data storage. Secure authentication is implemented using **JWT**, with AI functionalities powered by **OpenAI API** and **TensorFlow.js**. Translation is enabled via the **Google Cloud Translate API**, and the system is deployed using **Docker** and **Nginx** for scalability and reliability. [GitHub](#)

XAI Interpret

I developed **XAI Interpret**, a hands-on platform for model interpretability that implements state-of-the-art Explainable AI (XAI) techniques using SHAP and LIME. The project provides a complete machine learning pipeline, from data preprocessing and model training to real-time, interactive explanations for individual predictions. Supporting various ML algorithms, the system helps users understand both global and local model behavior, feature interactions, and produces rich visualizations for interpretability. Built for practical use, it's compatible with Google Colab and supports cloud deployment on Vertex AI, featuring production-ready model persistence, scalable architecture, and comprehensive LaTeX documentation. This solution fosters transparency and trust in machine learning models, making it especially valuable for black-box medical diagnostics and other high-stakes domains. [GitHub](#)

TECHNICAL SKILLS

- **Cloud Services:** AWS Console Management, Azure Portal Management, Google Cloud Platform
- **DevOps Tools:** Git/GitHub, Terraform, Redis, Docker, Kubernetes (Basics), GitLab
- **Programming Languages:** Java, Python, Dart, SQL, Shell Scripting
- **Development:** HTML, CSS, Tailwind CSS, JavaScript, Node.js, Express, EJS
- **CS Fundamentals:** DSA, DBMS, OS, CN
- **Databases:** MySQL, MongoDB
- **AI / ML & GenAI:** ML Fundamentals, Generative AI (LLMs), Prompt Engineering, AI Workflow Design, MLflow
- **Network Fundamentals:** Strong understanding of IP addressing, DNS, HTTP/HTTPS, and ports; capable of diagnosing connectivity and network-related issues.
- **Operating Systems:** Linux (Ubuntu, RHEL), Windows
- **Soft Skills:** Effective communication, Problem-solving, Time management, Critical thinking, Collaboration

CERTIFICATIONS

- **Fortinet Certified Fundamentals in Cybersecurity:** Fortinet
- **Fortinet Certified Associate in Cybersecurity:** Fortinet
- **Fundamentals of Cybersecurity (EDU-102):** Zscaler
- **Zero Trust Certified Associate (ZTCA):** Zscaler

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate:** Oracle
- **AWS Services for Solutions Architect Associate:** Udemy

ROLES AND RESPONSIBILITIES

Open Source Contributor — Open Source Contribution India

Aug 2025 - Sept 2025

Contributor

As an active open-source contributor with Open Source Contribution India, I have collaborated on community-driven software projects, engaging in code development, issue resolution, and peer reviews. My contributions have supported project growth and fostered innovation, while enhancing collaboration within the Indian open-source ecosystem.

Happier and Safer Internet Club — Cyber Security Awareness Club

Oct 2023 - Oct 2024

Core Member

As a founding member of the HSI Club, I actively participated in the UNESCO MIL 2023 Youth Hackathon and contributed to organizing debate competitions and cybersecurity awareness camps. I consistently promoted knowledge sharing about cyber-crimes and played a key role in spreading cybersecurity awareness within the community.

Matheletes — Maths Club

Mar 2023 - Jul 2024

Vice President

As a founding member of the Maths Club, I played a key role in designing and implementing innovative quiz formats and project display setups to engage participants. I actively collaborated with faculty and other student organizations to secure necessary resources, fostering a supportive environment for club activities. Leveraging digital tools for event planning and management, I ensured seamless logistics and provided real-time updates, greatly enhancing the experience for all participants.

LANGUAGES

- English
- Hindi
- Gujarati

AWARDS

- **SOF International Mathematics Olympiad:** School Rank: 3rd (2019)
- **Seamedu Awards 2025:** Creative Cloud Integration Award
- **IBM ICE Day Ideathon 2025:** Bronze Medal
- **Seamedu Awards 2026:** Multi Cloud Excellence Award

PUBLICATIONS

- **Title:** “Democratizing AWS Cloud Operations: A Unified Orchestration Approach To Standardized Infrastructure Management”
- Published in *International Journal of Creative Research Thoughts (IJCRT)*, Vol. 14, Issue 4, April 2026
- DOI: <https://doi.org/10.56975/ijcrt.v14i4.305033>
- Proposed Nebula, an AWS orchestration platform integrating provisioning, monitoring, and cost observability
- Designed asynchronous execution framework using Celery & Redis for scalable cloud operations